

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

Section 1 - Identification

Product Name Yeast Nitrogen Base

Product Code	R459952
Address	ThermoFisher Scientific Australia Pty Ltd 5 Caribbean Drive, Scoresby VICTORIA 3179, Australia
Emergency Tel.	CHEMTREC® 03 9757 4559 or +613 9757 4559
Telephone / Fax Numbers	Tel: 1300 735 292 Fax: 1800 067 639
E-mail address	ANZinfo@thermofisher.com

Recommended Use Laboratory chemicals.

Uses advised against This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

Section 2 - Hazard(s) Identification

Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

Physical hazards

No hazards identified

Health hazards

Skin Corrosion/Irritation	Category 2
Serious Eye Damage/Eye Irritation	Category 2
Specific target organ toxicity - (single exposure)	Category 3

Environmental hazards

No hazards identified

Label Elements



Exclamation Mark

Signal Word**Warning****Hazard Statements**

H315 - Causes skin irritation
 H319 - Causes serious eye irritation
 H335 - May cause respiratory irritation

Precautionary Statements

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray
 P264 - Wash face, hands and any exposed skin thoroughly after handling
 P271 - Use only outdoors or in a well-ventilated area
 P280 - Wear protective gloves/protective clothing/eye protection/face protection
 P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
 P312 - Call a POISON CENTER or doctor if you feel unwell
 P362 + P364 - Take off contaminated clothing and wash it before reuse
 P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
 P501 - Dispose of contents/ container to an approved waste disposal plant

Other information

Toxic to terrestrial vertebrates
 This product does not contain any known or suspected endocrine disruptors

Section 3 - Composition and Information on Ingredients

Component	CAS No	Weight %
Sodium carbonate	497-19-8	0.59
Histidine, L-, monohydrochloride monohydrate	5934-29-2	0.15
DL-Methionine	59-51-8	0.3
DL-Tryptophan	54-12-6	0.3
Ammonium sulfate	7783-20-2	73.63
Magnesium sulfate	7487-88-9	7.26
Calcium chloride	10043-52-4	1.45
Citric acid	77-92-9	0.29
Dihydrogen potassium phosphate	7778-77-0	14.51
Sodium molybdate dihydrate	10102-40-6	Trace
Riboflavin	83-88-5	Trace
Iron(III) chloride	7705-08-0	Trace
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	67-03-8	Trace
Zinc sulfate heptahydrate	7446-20-0	Trace
3-Pyridinecarboxylic acid	59-67-6	Trace
Potassium iodide	7681-11-0	Trace
Copper (II) sulfate pentahydrate (1:1:5)	7758-99-8	Trace
L-Glutamic acid, N-[4-[(2-amino-1,4-dihydro-4-oxo-6-pteridiny)l)methyl]aminobenzoyl]-	59-30-3	Trace
(+)-Biotin	58-85-5	Trace
Benzoic acid, 4-amino-, monosodium salt	555-06-6	Trace
Boric acid (H3BO3)	10043-35-3	Trace
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	58-56-0	Trace
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	137-08-6	Trace
Sodium chloride	7647-14-5	1.45
myo-Inositol	87-89-8	Trace

Section 4 - First Aid Measures

Inhalation	Remove to fresh air.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.
Eye Contact	Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.
First Aid Facilities	Eyewash, safety shower and washroom.
Most important symptoms and effects	No information available.
Notes to Physician	Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Section 6 - Accidental Release Measures

Emergency procedures

Ensure adequate ventilation.

Environmental Precautions

See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

Methods for Containment and Clean Up

Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal.

Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

Section 7 - Handling and Storage

Precautions for Safe Handling

Ensure adequate ventilation.

Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

Section 8 - Exposure Controls and Personal Protection

Exposure limits

AUS - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

ACGIH - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

UK - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

Component	Australia	New Zealand WEL	ACGIH TLV	The United Kingdom	Germany
Citric acid					TWA: 2 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 mg/m ³
Sodium molybdate dihydrate	TWA: 5 mg/m ³		TWA: 0.5 mg/m ³	STEL: 10 mg/m ³ 15 min TWA: 5 mg/m ³ 8 hr	
Iron(III) chloride	TWA: 1 mg/m ³		TWA: 1 mg/m ³	STEL: 2 mg/m ³ 15 min TWA: 1 mg/m ³ 8 hr	
Zinc sulfate heptahydrate					TWA: 0.1 mg/m ³ (8 Stunden). MAK TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.4 mg/m ³ Höhepunkt: 4 mg/m ³
Potassium iodide			TWA: 0.01 ppm		
Copper (II) sulfate pentahydrate (1:1:5)			TWA: 1 mg/m ³	STEL: 2 mg/m ³ 15 min TWA: 1 mg/m ³ 8 hr	TWA: 0.01 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.02 mg/m ³
Boric acid (H3BO3)			TWA: 2 mg/m ³ STEL: 6 mg/m ³		TWA: 0.5 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 mg/m ³ (8 Stunden). MAK when boric acid and tetraborates are present together, the MAK value is 0.75 mg boron/m ³ Höhepunkt: 10 mg/m ³

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Exposure Controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment**Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	AUS/NZ Standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	AS/NZS 2161	(minimum requirement)

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Skin and body protection

Long sleeved clothing

Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)
When RPE is used a face piece Fit Test should be conducted

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties**Appearance****Physical State**

Solid

Odor

No information available

Odor Threshold

No data available

pH

No information available

Melting Point/Range

No data available

Softening Point

No data available

Boiling Point/Range

No information available

Flash Point

No information available

Evaporation Rate

Not applicable

Method - No information available
Solid

Flammability (solid,gas)

No information available

Explosion Limits

No data available

Vapor Pressure

No data available

Vapor Density

Not applicable

Solid

Specific Gravity / Density

No data available

Bulk Density

No data available

Water Solubility

No information available

Solubility in other solvents

No information available

Partition Coefficient (n-octanol/water)

Component	log Pow	
Ammonium sulfate	-5.1	
Citric acid	-1.72	
Riboflavin	-1.46	
Iron(III) chloride	-4	
Thiazolium,	-3.04	
3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl-chloride, monohydrochloride		
3-Pyridinecarboxylic acid	2.34	
Potassium iodide	0.04	
L-Glutamic acid,	-2.41	
N-[4-[[[(2-amino-1,4-dihydro-4-oxo-6-pyridinyl)methyl]amino]benzoyl]-Boric acid (H ₃ BO ₃)	-0.757	
3,4-Pyridinedimethanol,	-0.7	
5-hydroxy-6-methyl-, hydrochloride		
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	-5.35	
myo-Inositol	-2.08	
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	Not applicable	Solid
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other information

Section 10 - Stability and Reactivity

Reactivity	None known, based on information available
Stability	Stable under normal conditions.
Conditions to Avoid	Heat, flames and sparks.
Incompatible Materials	None known.
Hazardous Decomposition Products	None under normal use conditions.
Hazardous Polymerization	No information available.

Section 11 - Toxicological Information

Information on Toxicological Effects**Product Information****(a) acute toxicity;**

Oral	Based on available data, the classification criteria are not met
Dermal	Based on available data, the classification criteria are not met
Inhalation	Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Sodium carbonate	2800 mg/kg (Rat)	> 2000 mg/kg (rabbit)	2.3 mg/l 2h (Rat)
Ammonium sulfate	LD50 = 2840 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	

Calcium chloride	2301 mg/kg (Rat)	LD50 > 5000 mg/kg (Rabbit)	
Citric acid	LD50 = 3 g/kg (Rat)	>2 g/kg (Rat)	
Dihydrogen potassium phosphate	LD50 = 3200 mg/kg (Rat)	LD50 > 4640 mg/kg (Rabbit)	LC50 > 0.83 mg/L (Rat) 4 h
Riboflavin	LD50 > 10 g/kg (Rat)		>5.4 mg/L/4h (rat)
Iron(III) chloride	450 mg/kg (Rat) 316 mg/kg (Rat)		
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LD50 = 3710 mg/kg (Rat)		
Zinc sulfate heptahydrate	1260 mg/kg (Rat)		
3-Pyridinecarboxylic acid	LD50 = 7 g/kg (Rat)	LD50 > 2000 mg/kg (Rat)	LC50 > 3.8 mg/L (Rat) 4 h
Potassium iodide	2779 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	
Copper (II) sulfate pentahydrate (1:1:5)	LD50 = 960 mg/kg (Rat)	LD50 > 8 g/kg (Rabbit)	
L-Glutamic acid, N-[4-[[[(2-amino-1,4-dihydro-4-oxo-6-pteridin yl)methyl]amino]benzoyl]-	10 g/kg (Mouse)		
Boric acid (H3BO3)	2660 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	Not listed
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	4 g/kg (Rat)		
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)- beta.-alanine, calcium salt (2:1), (R)-	LD50 > 10 g/kg (Rat)		
Sodium chloride	LD50 = 3 g/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable
Solid

Symptoms / effects, both acute and delayed No information available

Section 12 - Ecological Information

Ecotoxicity effects

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Sodium carbonate	Lepomis macrochirus: LC50: 300 mg/L/96h Gambusia affinis: LC50: 740 mg/L/96h	EC50: = 265 mg/L, 48h (Daphnia magna)		-
Ammonium sulfate	LC50: = 18 mg/L, 96h (Cyprinus carpio) LC50: 32.2 - 41.9 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: 5.2 - 8.2 mg/L, 96h static (Oncorhynchus mykiss) LC50: > 100 mg/L, 96h (Pimephales promelas) LC50: 123 - 128 mg/L, 96h semi-static (Poecilia reticulata) LC50: = 126 mg/L, 96h (Poecilia reticulata) LC50: = 420 mg/L, 96h semi-static (Brachydanio rerio) LC50: = 250 mg/L, 96h (Brachydanio rerio) LC50: = 480 mg/L, 96h flow-through (Brachydanio rerio)	LC50: = 14 mg/L, 48h (Daphnia magna)		
Magnesium sulfate	LC50: 2610 - 3080 mg/L, 96h static (Pimephales promelas)	EC50: 266.4 - 417.3 mg/L, 48h Static (Daphnia magna)	EC50: = 2700 mg/L, 72h (Desmodesmus subspicatus)	= 84000 mg/L EC50 Photobacterium phosphoreum 30 min
Calcium chloride	Lepomis macrochirus: LC50: 10650 mg/L/96h	EC50: 52 mg/L/48h		
Citric acid	Leuciscus idus: LC50 = 440-760 mg/L/96h	EC50 = 120 mg/L/72h		Photobacterium phosphoreum: EC50 = 14 mg/L/15 min
Riboflavin	Brachydanio rerio: LC50 >10000 mg/L/96h	EC50 >7.4 mg/L/48h		
Iron(III) chloride	LC50: 20.95 - 22.56 mg/L, 96h semi-static (Pimephales promelas) LC50: = 20.26 mg/L, 96h semi-static (Lepomis macrochirus)	EC50: = 9.6 mg/L, 48h Static (Daphnia magna) EC50: = 27.9 mg/L, 48h (Daphnia magna)		
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LC50 >100 mg/L/96h	EC50 >100 mg/L/48h		
Zinc sulfate heptahydrate	1.9 mg/L LC50 96 h			
3-Pyridinecarboxylic acid	LC50: = 520 mg/L, 96h (Salmo trutta)	EC50: = 77 mg/L, 48h (Daphnia magna)	EC50: = 89.93 mg/L, 72h (Desmodesmus subspicatus)	= 160 mg/L EC50 Salmonella typhimurium 72 h = 2792.91 mg/L EC50 Tetrahymena pyriformis 60 h
Potassium iodide	Onchorhynchus mykiss: LC50: 3200 mg/L/120h	-	-	-
Copper (II) sulfate pentahydrate (1:1:5)	Onchorhynchus mykiss: LC50 = 0.1-2.5	EC50 = 0.24 mg/L/48h		Photobacterium phosphoreum: EC50 =

	mg/L/96h			0.25 mg/L/30min as Cu++ Photobacterium phosphoreum EC50= 1.3 mg/L/5 min as Cu++
Boric acid (H3BO3)	Gambusia affinis: LC50: 5600 mg/L/96h	EC50: 115 - 153 mg/L, 48h (Daphnia magna)	-	-
Sodium chloride	Pimephals prome: LC50: 7650 mg/L/96h	EC50: 1000 mg/L/48h		

Persistence and Degradability	No information available
Degradation in sewage treatment plant	Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.
Bioaccumulative Potential	No information available

Component	log Pow	Bioconcentration factor (BCF)
Ammonium sulfate	-5.1	No data available
Citric acid	-1.72	No data available
Riboflavin	-1.46	No data available
Iron(III) chloride	-4	2756 - 9622 dimensionless
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	-3.04	No data available
3-Pyridinecarboxylic acid	2.34	No data available
Potassium iodide	0.04	No data available
L-Glutamic acid, N-[4-[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	-2.41	No data available
Boric acid (H3BO3)	-0.757	0 dimensionless
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	-0.7	No data available
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-beta.-alanine, calcium salt (2:1), (R)-myo-Inositol	-5.35	No data available
	-2.08	No data available

Mobility	No information available.
Endocrine Disruptor Information	This product does not contain any known or suspected endocrine disruptors
Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

Section 13 - Disposal Considerations

Waste from Residues/Unused Products	Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point.
Other Information	Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

Section 14 - Transport Information

IMDG/IMO Not regulated

ADG Not regulated

Component	Hazchem Code
Iron(III) chloride	2X

7705-08-0 (Trace)

<u>IATA</u>	Not regulated
Environmental hazards	No hazards identified
Special Precautions	No special precautions required
Additional information	None known

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations **Australia**

See section 8 for national exposure control parameters.

Standard for the Uniform Scheduling of Medicines and Poisons

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

Component	Standard for the Uniform Scheduling of Medicines and Poisons
Sodium carbonate - 497-19-8	Schedule 5 listed - being the carbonate, silicate or phosphate salts of sodium or potassium alone or in any combination: in solid orthodontic device cleaning preparations, the pH of which as an in-use aqueous solution is >11.5, in solid automatic dishwashing preparations, the pH of which in a 500 g/L aqueous solution or mixture is >11.5 but <=12.5; in other solid preparations, the pH of which in a 10 g/L aqueous solution is >11.5, or in liquid or semi-solid preparations, the pH of which is >11.5, unless: in food additive preparations for domestic use, or in automatic dish washing preparations for domestic use with a pH >12.5; except when separately specified in these Schedules Schedule 5 listed - being the Carbonate, Silicate or Phosphate salts of Sodium or Potassium alone or in any combination: in solid orthodontic device cleaning preparations as an in-use aqueous solution; in solid automatic dishwashing preparations in a 500 g/L aqueous solution or mixture but with pH <=12.5; in other solid preparations in a 10 g/L aqueous solution, or in liquid or semi-solid preparations, unless in food additive preparations for domestic use, or in automatic dish washing preparations for domestic use with a pH >12.5; except when separately specified in these Schedules Schedule 6 listed - being the carbonate, silicate or phosphate salts of sodium or potassium alone or in any combination for non-domestic use: in solid automatic dishwashing preparations, the pH of which in a 500 g/L aqueous solution or mixture is >12.5, or in liquid or semi-solid automatic dishwashing preparations, the pH of which is >12.5 Schedule 10 listed
DL-Tryptophan - 54-12-6	Schedule 4 listed - for human therapeutic use except in preparations labelled with a recommended daily dose of <=100 mg of Tryptophan
Magnesium sulfate - 7487-88-9	Schedule 3 listed
Dihydrogen potassium phosphate - 7778-77-0	Schedule 10 listed
Iron(III) chloride - 7705-08-0	Schedule 2 listed Schedule 4 listed - in injectable preparations for human use Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient; in preparations for injection except in preparations containing <=0.1% of Iron Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient; in other preparations except in liquid or gel preparations containing <=0.1% of Iron, or in animal feeds or feed premixes Schedule 5 listed - for use as agricultural chemicals except in preparations containing <=4% of Iron Schedule 6 listed - except up to 1% of Iron oxides when present as an excipient. For the treatment of animals except: when included in Schedule 5, in liquid or gel preparations containing <=0.1% of Iron, or in animal feeds or feed premixes
Zinc sulfate heptahydrate - 7446-20-0	Schedule 4 listed - for human internal use except in preparations with a recommended daily dose of <=25 mg of Zinc, or in preparations with a recommended daily dose of between 25-50 mg of Zinc when compliant with the requirements of the Required Advisory Statements for Medicine Labels
3-Pyridinecarboxylic acid - 59-67-6	Schedule 3 listed Schedule 4 listed - for human therapeutic use except: when contained in other schedules, in preparations containing <=100 mg of Nicotinic acid per dosage unit, or Nicotinamide

Copper (II) sulfate pentahydrate (1:1:5) - 7758-99-8	Schedule 4 listed - for human use except: when separately specified in these Schedules, in preparations for human internal use containing ≤5 mg of Copper per recommended daily dose, or in other preparations containing ≤5% of Copper compounds Schedule 5 listed - in animal feed additives except in preparations containing ≤1% of Copper Schedule 6 listed - except: when separately specified in these Schedules, in preparations for human internal use containing ≤5 mg of Copper per recommended daily dose, pigments where the solubility of the Copper compounds in water is ≤1 g/L, in feed additives containing ≤1% of Copper, or in other preparations containing ≤5% of Copper compounds Schedule 6 listed - except when separately specified in these Schedules; in preparations for human internal use containing ≤5 mg of Copper per recommended daily dose; pigments where the solubility of the Copper compounds in water is ≤1 g/L; in feed additives containing ≤1% of Copper, or in other preparations containing ≤5% of Copper compounds
L-Glutamic acid, N-[4-[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl] - 59-30-3	Schedule 2 listed Schedule 4 listed - in preparations for human use for injection
Boric acid (H ₃ BO ₃) - 10043-35-3	Schedule 4 listed - for human therapeutic use except: in preparations for internal use containing ≤6 mg Boron per recommended daily dose, in preparations for dermal use containing ≤0.35% of Boron, which are not for paediatric or antifungal use, or when present as an excipient Schedule 5 listed - except: a) when included in Schedule 4, or b) in cosmetic hand cleaning preparations when labelled with a warning to the following effect: NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, and if the concentration of free soluble Borates >1.5% (as Boric acid), with the words: NOT TO BE USED ON PEELING OR IRRITATED SKIN, or c) in cosmetic Talc preparations containing ≤5% calculated as Boric acid when labelled with a warning to the following effect: NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, and if the concentration of free soluble Borates >1.5% (as Boric acid), with the words: NOT TO BE USED ON PEELING OR IRRITATED SKIN, or d) in cosmetic oral hygiene preparations containing ≤0.1% calculated as Boric acid when labelled with a warning to the following effect: NOT TO BE SWALLOWED. NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, or e) in other cosmetic preparations containing ≤3% calculated as Boric acid when labelled with a warning to the following effect: NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, and if the concentration of free soluble Borates >1.5% (as Boric acid), with the words: NOT TO BE USED ON PEELING OR IRRITATED SKIN, or f) in preparations, other than insect baits, containing ≤6%, calculated as Boric acid

Australian Industrial Chemicals Introduction Scheme (AICIS)

Component	Australian Industrial Chemicals Introduction Scheme (AICIS)	Additional information
Sodium carbonate - 497-19-8	Present	-
Histidine, L-, monohydrochloride monohydrate - 5934-29-2	Present	-
DL-Methionine - 59-51-8	Present	-
DL-Tryptophan - 54-12-6	Present	-
Ammonium sulfate - 7783-20-2	Present	-
Magnesium sulfate - 7487-88-9	Present	-
Calcium chloride - 10043-52-4	Present	-
Citric acid - 77-92-9	Present	-
Dihydrogen potassium phosphate - 7778-77-0	Present	-
Sodium molybdate dihydrate - 10102-40-6	Present	-
Riboflavin - 83-88-5	Present	-
Iron(III) chloride - 7705-08-0	Present	-
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride - 67-03-8	Present	-
Zinc sulfate heptahydrate - 7446-20-0	Present	-
3-Pyridinecarboxylic acid - 59-67-6	Present	-
Potassium iodide - 7681-11-0	Present	-
Copper (II) sulfate pentahydrate (1:1:5) - 7758-99-8	Present	-
L-Glutamic acid, N-[4-[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl] - 59-30-3	Present	-
(+)-Biotin - 58-85-5	Present	-
Benzoic acid, 4-amino-, monosodium salt - 555-06-6	Present	-
Boric acid (H ₃ BO ₃) - 10043-35-3	Present	-
3,4-Pyridinedimethanol,	Present	-

5-hydroxy-6-methyl-, hydrochloride - 58-56-0		
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-beta.-alanine, calcium salt (2:1), (R)- - 137-08-6	Present	-
Sodium chloride - 7647-14-5	Present	-
myo-Inositol - 87-89-8	Present	-

Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

National pollutant inventory Not applicable

Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

International Inventories

Component	AICS	NZIoC	EINECS	ELINCS	TSCA	DSL	NDSL	PICCS	ENCS	ISHL	IECSC	KECL
Sodium carbonate	X	X	207-838-8	-	X	X	-	X	X	X	X	KE-31380
Histidine, L-, monohydrochloride monohydrate	X	X	-	-	-	-	-	X	-		X	-
DL-Methionine	X	X	200-432-1	-	X	X	-	X	X	X	X	KE-14-0064
DL-Tryptophan	X	X	200-194-9	-	X	X	-	X	X	X	-	2014-3-5933
Ammonium sulfate	X	X	231-984-1	-	X	X	-	X	X	X	X	KE-01743
Magnesium sulfate	X	X	231-298-2	-	X	X	-	X	X	X	X	KE-22752
Calcium chloride	X	X	233-140-8	-	X	X	-	X	X	X	X	KE-04496
Citric acid	X	X	201-069-1	-	X	X	-	X	X	X	X	KE-20831
Dihydrogen potassium phosphate	X	X	231-913-4	-	X	X	-	X	X	X	X	KE-28622
Sodium molybdate dihydrate	X	X	-	-	-	-	-	X	X	X	X	-
Riboflavin	X	X	201-507-1	-	X	X	-	X	X	X	X	KE-11845
Iron(III) chloride	X	X	231-729-4	-	X	X	-	X	X	X	X	KE-21134
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	X	X	200-641-8	-	X	X	-	X	X	X	X	KE-01482
Zinc sulfate heptahydrate	X	X	-	-	-	X	-	X	X		X	-
3-Pyridinecarboxylic acid	X	X	200-441-0	-	X	X	-	X	X	X	X	KE-29937
Potassium iodide	X	X	231-659-4	-	X	X	-	X	X	X	X	KE-29149
Copper (II) sulfate pentahydrate (1:1:5)	X	X	-	-	-	-	-	X	X		X	-
L-Glutamic acid, N-[4-[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	X	X	200-419-0	-	X	X	-	X	X		X	KE-01305
(+)-Biotin	X	X	200-399-3	-	X	X	-	X	X	X	X	KE-18590
Benzoic acid,	X	-	209-080-3	-	X	-	X	-	X	X	X	-

4-amino-, monosodium salt												
Boric acid (H ₃ BO ₃)	X	X	233-139-2	-	X	X	-	X	X	X	X	KE-03499
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	X	X	200-386-2	-	X	X	-	X	X	X	X	KE-20695
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-beta.-alanine, calcium salt (2:1), (R)-	X	X	205-278-9	-	X	X	-	X	X	X	X	KE-10821
Sodium chloride	X	X	231-598-3	-	X	X	-	X	X	X	X	KE-31387
myo-Inositol	X	X	201-781-2	-	X	X	-	X	X	X	X	KE-21013

Legend: X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

Component	Basel Convention (Hazardous Waste)	Australian Hazardous Waste Act - Categories of Wastes to Be Controlled
Citric acid - 77-92-9	Annex I - Y34	Y34 solid or solution
Zinc sulfate heptahydrate - 7446-20-0	Annex I - Y23	Y23
Copper (II) sulfate pentahydrate (1:1:5) - 7758-99-8	Annex I - Y22	Y22

Component	CAS No	OECD HPV	Restriction of Hazardous Substances (RoHS)	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Sodium carbonate	497-19-8	Listed	Not applicable	Not applicable	Not applicable
Histidine, L-, monohydrochloride monohydrate	5934-29-2	Not applicable	Not applicable	Not applicable	Not applicable
DL-Methionine	59-51-8	Listed	Not applicable	Not applicable	Not applicable
DL-Tryptophan	54-12-6	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium sulfate	7783-20-2	Listed	Not applicable	Not applicable	Not applicable
Magnesium sulfate	7487-88-9	Listed	Not applicable	Not applicable	Not applicable
Calcium chloride	10043-52-4	Listed	Not applicable	Not applicable	Not applicable
Citric acid	77-92-9	Listed	Not applicable	Not applicable	Not applicable
Dihydrogen potassium phosphate	7778-77-0	Listed	Not applicable	Not applicable	Not applicable
Sodium molybdate dihydrate	10102-40-6	Not applicable	Not applicable	Not applicable	Not applicable
Riboflavin	83-88-5	Listed	Not applicable	Not applicable	Not applicable
Iron(III) chloride	7705-08-0	Listed	Not applicable	Not applicable	Not applicable
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	67-03-8	Not applicable	Not applicable	Not applicable	Not applicable
Zinc sulfate heptahydrate	7446-20-0	Not applicable	Not applicable	Not applicable	Not applicable

3-Pyridinecarboxylic acid	59-67-6	Listed	Not applicable	Not applicable	Not applicable
Potassium iodide	7681-11-0	Listed	Not applicable	Not applicable	Not applicable
Copper (II) sulfate pentahydrate (1:1:5)	7758-99-8	Listed	Not applicable	Not applicable	Not applicable
L-Glutamic acid, N-[4-[[[(2-amino-1,4-dihydro-4-oxo-6-pteridiny)l)methyl]amino]benzoyl]-	59-30-3	Not applicable	Not applicable	Not applicable	Not applicable
(+)-Biotin	58-85-5	Not applicable	Not applicable	Not applicable	Not applicable
Benzoic acid, 4-amino-, monosodium salt	555-06-6	Not applicable	Not applicable	Not applicable	Not applicable
Boric acid (H3BO3)	10043-35-3	Listed	Not applicable	Not applicable	Not applicable
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	58-56-0	Listed	Not applicable	Not applicable	Not applicable
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	137-08-6	Not applicable	Not applicable	Not applicable	Not applicable
Sodium chloride	7647-14-5	Listed	Not applicable	Not applicable	Not applicable
myo-Inositol	87-89-8	Not applicable	Not applicable	Not applicable	Not applicable

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Sodium carbonate	-	Use restricted. See item 75. (see link for restriction details)	-
Ammonium sulfate	-	Use restricted. See item 65. (see link for restriction details)	-
Calcium chloride	-	Use restricted. See item 75. (see link for restriction details)	-
Citric acid	-	Use restricted. See item 75. (see link for restriction details)	-
Riboflavin	-	Use restricted. See item 75. (see link for restriction details)	-
Zinc sulfate heptahydrate	-	Use restricted. See item 75. (see link for restriction details)	-
Copper (II) sulfate pentahydrate (1:1:5)	-	Use restricted. See item 75. (see link for restriction details)	-
Boric acid (H3BO3)	-	Use restricted. See item 30. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 233-139-2 - Toxic for reproduction, Article 57c

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

Section 16 - Other Information

Legend

AICS - Australian Inventory of Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

NZIoC - New Zealand Inventory of Chemicals

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists
Predicted No Effect Concentration (PNEC)

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

NZS 5433:2020 - Transport of Dangerous Goods on Land

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

OECD - Organisation for Economic Co-operation and Development

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Revision Date 05-Jul-2023

Revision Summary Not applicable.

This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet